

Which Hand-Motion Composition Axes Are Worth Measuring

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I. INTRODUCTION

Motion priors for dexterous hands are usually monolithic, emitting one continuous latent per control step. LAMP [1] is the current instance for hands, and it rejects a quantized alternative because “codebook switches can turn small latent changes into abrupt hand-command jumps during contact-rich residual RL”. Whole-body animation went the other way: MotionBricks [2] builds a modular latent backbone over more than 350,000 clips, but a human composes its primitives through an authoring interface, so it does not test whether the units of a *learned* bank mean anything stable.

Both architectures assume that separately learned primitives recombine, and that novel recombinations stay usable. Does modularity buy that at the scale hand datasets offer, when MotionBricks trains on 10^5 clips and the largest public hand dataset here provides 10^3 ? And does any of it reach a robot, when a composition can be valid human motion and still unreachable on a hand whose fingers are spaced differently? We build a measurement for the first and report a null for the second.

Our contributions are as follows.

- 1) A paired measurement of compositional difficulty, with the five gates that make its output trustworthy (§III-C, Table I). Four exist because an earlier design returned plausible numbers it could not have earned.
- 2) The finding that compositional difficulty needs the two factors to interact, and that informative factors alone do not produce it. The relation holds across six axes on three datasets, but only one could have

falsified the necessity direction, so we state it as one-sided evidence. A statistic finds the interaction without training anything and diagnoses presence well. It ranks badly, however, since the one axis we swept on its recommendation came back the hardest of the six (§V-B).

- 1) A null on the hardware question, and an account of why it is a weak null. The retargeting our transfer rests on has no collision term, the banks behind it were trained on the one dataset our own instrument finds null, and the equivalence test that would license “the same” fails its own sensitivity check.

That transfer is our measuring apparatus and not our contribution: work carrying human hand-object motion onto real hardware [7], [13] is further along that axis than this paper. We adopt all three architectures. The switching formulation is rSLDS [3] and SNLDS [4], and the weight-blending control is MANN [5], whose gating network “dynamically updates

the weights of the motion prediction network by selecting and blending what we call the expert weights”.

II. RELATED WORK

LAMP [1] learns a history-conditioned continuous latent prior over hand actions and uses it as the action space for real-world residual reinforcement learning. DexSynRefine [7] represents human hand-object motion with motion-manifold primitives and grounds it in physics with task-space residual RL. Neither evaluates whether a prior’s primitives generalize to combinations absent from training. Our parameter-matched baseline *is* a LAMP-style continuous latent, and our modular arm takes its backbone from MotionBricks [2], whose primitives a human authors through an interface. We study the learned bank that replaces them, and claim it as an object of study only.

Modular motion models recur in mechanically distinct forms, from an authored-primitive latent backbone [2] to switching linear [3] and nonlinear [4] dynamical systems and mode-adaptive gating [5], all sharing the premise that separately learned primitives can be recombined. A survey of over 700 movement-primitive papers [8] lists composing primitives and evaluating frameworks among its open questions.

Recent work uses “compositional” in senses that do not answer that question. CoDex [9] composes the *stages* of a task and learns no motion prior; UniCross [10] composes four hand-defined skills and generalizes to unseen *objects*; object-centric neural fields [11] mix object-conditioned primitives and also evaluate on unseen *objects*. In each case the held-out axis is the object or the task stage, and none holds out the pairing of two factors.

Adversarial Concept Search [12] predicts which concept combinations a model will fail on from the geometry of its learned representations. It is the closest work to our screen, and it needs a trained model and works on language, where §III-D’s statistic is computed on raw data factors before any model exists.

SPIDER [13] converts kinematic human demonstrations into dynamically feasible robot trajectories at scale. We claim no contribution to retargeting, and §VI states why the transfer used here is weaker than the field’s standard.

III. METHOD

A. Hand representation

Motion is 27 anatomical degrees of freedom in native units: four per finger (MCP flexion and abduction, PIP, DIP),

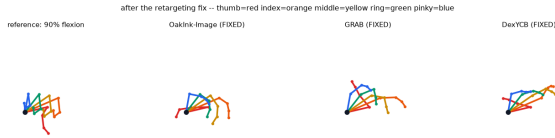


Fig. 1. The same 90%-flexion reference pose projected onto the anatomical model from each dataset, after the conventions are inferred per joint. Thumb red, index orange, middle yellow, ring green, little blue. The three datasets agree on the pose; before the fix, an assumed abduction axis put 100% of GRAB and DexYCB frames outside `thumb_mcp_abd`’s range.

five for the thumb, six for the wrist. Normalization into $[-1, 1]$ comes from the joint-limit box, never from dataset statistics, which keeps a prior trained on one source meaningful on another.

Public hand datasets provide MANO [20] parameters, and projecting them onto an anatomical model requires three conventions: which Euler axis carries flexion, its sign, and which of the remaining two carries abduction. We infer all three per joint and assume none. We did assume them first, and the thumb and the fingers turned out not to follow each other’s convention.¹ Assuming costs in both directions, and we paid both. Take abduction as $(\text{flexion} + 1) \bmod 3$, correct for the fingers, and 100% of GRAB and DexYCB frames fall outside `thumb_mcp_abd`’s 20° range. Pool PIP variance across fingers instead, and their differing resting offsets count as flexion; on OakInk2 that selected the wrong axis and pinned `index_pip_flex` at its limit in every frame. Inferring per joint halves that dataset’s projection residual and takes its PIP joints from 37.3% inside their anatomical range to 99.9%.

We count usable DOF out of 27 before and after these corrections. Dominance is the flexion axis’s share of joint variance, and the residual is the mean per-joint angular error, in degrees, of reprojecting the anatomical DOF back to MANO’s axis-angle parameters. The count exempts the three wrist-rotation channels, which are constant zero in every MANO-derived bundle and would otherwise register as dead, so 26 of 27 means 23 of the 24 MANO can express. That is not a defect: MANO has no forearm, so its root rotation gives the hand’s absolute orientation in the capture frame, which is not an articulation, and filling those channels would encode where the camera stood. Every error in this paper is averaged over 27 channels, three of them identically zero in both arms. No released artifact records a before usable-DOF count for any dataset. Those four figures come from the project’s notes. OakInk2’s and GRAB’s before dominances are in the rebuild logs. OakInk-Image’s and DexYCB’s come from the same notes and are likewise unreleased.

¹MANO’s zero pose is a flat hand chosen for compatibility with SMPL [20], and its release carries a kinematic tree with no per-joint rest rotations, so every joint frame is parallel to the template’s. The thumb’s is therefore no better aligned to its own anatomy than a finger’s, and its flexion and abduction fall on different Euler components.

dataset	usable DOF	dominance	residual
OakInk-Image	23 $\rightarrow$ 26	0.663 $\rightarrow$ 0.726	6.01 $^\circ$
OakInk2	19 $\rightarrow$ 23	0.470 $\rightarrow$ 0.571	21.86 $^\circ \rightarrow$ 10.22°
DexYCB	19 $\rightarrow$ 23	0.470 $\rightarrow$ 0.537	12.26 $^\circ$
GRAB	16 $\rightarrow$ 21	0.450 $\rightarrow$ 0.533	16.15 $^\circ$

B. The two priors

Architecture. Both arms encode a 32-frame window of 27 anatomical DOF and reconstruct it under a bounded output, which makes joint-limit satisfaction architectural. The modular arm emits a relaxed categorical assignment over a bank of K primitives, one per frame, each a 32-dimensional embedding, plus one 4-D style code for the window, kept small so it cannot let the decoder bypass the bank. The assignment uses Gumbel-softmax [22] in training and a plain softmax at evaluation, so the reported numbers pass through a convex combination and never a hard switch. The baseline replaces the bank with a 12-D Gaussian latent, also per frame. Both encode with bidirectional two-layer GRUs of 256 units and decode with unidirectional ones. Section V-A’s sweeps use $K = 12$, where the baseline has 2,283,059 parameters and the modular arm 2,295,727, a difference of 0.6%; §V-D and §V-C’s pooled banks use $K = 16$ (2,297,907); §V-C’s two weight sweeps use $K = 12$.

Objective. Both are trained as beta-VAEs [23] on windowed reconstruction, with mean-squared error on pose, a first-difference term because the data is 30 fps motion, and β times a KL with a free-bits floor of 0.02 nats per dimension. The weight β is annealed over the first epochs and validation is always scored at its final value. The modular arm adds four terms. They shape the bank, not the reconstruction: a switch penalty between adjacent frames, a per-frame entropy term, a usage-entropy reward and a load-balance term.

A fifth term is optional and off in §V-A’s sweeps. For each primitive it decodes that primitive after several predecessors and from several initial poses and penalizes the spread of the resulting velocity profiles. Velocity, because pose is dominated by the initial condition. It optimizes the quantity §V-C measures, which is why §V-C reads its own number as a target being met.

Rate matching. The quantity to match is the rate of information flow, which the parameter count does not fix. A window-level latent would push 32 frames through one code while the bank emits an assignment every frame, so the baseline gets a per-frame latent too. That equalizes how often a code is emitted, and on nominal capacity it leaves the baseline ahead: a 12-D continuous Gaussian carries more per frame than a 12-way categorical whose 4-D style code is paid once per window. On realized rate the arms are not matched at all, and the gap favors the modular arm. Rates below are final-epoch validation quantities in nats per frame, with assignment rate estimated as usage entropy minus assignment entropy.

Is the baseline’s KL binding? The asymmetry is one of penalty: the baseline’s latent is charged a KL and the assignment channel is charged nothing. That

sweep		n models	baseline rate	modular assignment rate	ratio
confirmatory axis, original sweep	category	140 / 139	0.207	2.275	11.0×
pc_hard, the positive control, both budgets		46 / 46	0.223	2.399	10.8×
runs/pc_easy at budget 256, the negative control		10 / 10	0.230	2.424	10.6×

penalty is active and binding across the three sweeps above. `latent_prior.py` clamps per dimension, and `kl_clamped` exceeds its $12 \times 0.02 = 0.24$ floor on the training split in 102 of 140, 45 of 46 and 19 of 20 baselines. The 19 of 20 is taken over both `pc_easy` budgets, not the budget-256 row above. The rate-control re-sweep is weaker on both counts, at 92 of 140 and a per-model minimum of zero active units, with three unmatched baselines fully collapsed; excluding the six seeds below 0.9 active units moves its unmatched advantage from +0.00693 to +0.00649. Nor is the latent collapsed, with mean active units between 0.996 and 1.000 across the three sweeps and a per-model minimum of 0.833. On the other side, a low conditional assignment entropy of 0.160 nats against a $\ln 12 = 2.485$ maximum means the selection is nearly deterministic given the input, which makes it a high-rate channel.

The matched arm. Raising the baseline’s free-bits floor to 0.186 per dimension closes the gap. Its realized KL rises to 1.846 nats per frame against the modular arm’s 2.312, a ratio of 1.25 where the unmatched baseline in the same sweep sits at 0.206 for a ratio of 11.2. The confirmatory axis was re-swept on that basis, with all three arms trained together over the same 70 declared seeds, all of which returned every arm.

That intervention produces a result, so §V-A reports it. One sensitivity check belongs to the intervention itself. Five of the 70 seeds cover 4 of 5 held compositions. Excluding them moves the raw figure to +0.00273 at $p = 0.052$, so they carry neither reading. One caution applies to any comparison drawn across sweeps. §VI gives it.

C. Paired compositional measurement

We score the same target trajectories under two priors, differing only in whether their training data contained the target’s composition:

$$\text{penalty} = \text{MSE}_{\text{naive}}(T) - \text{MSE}_{\text{informed}}(T) \text{ paired} = \text{penalty}_{\text{baseline}} - \text{penalty}_{\text{modular}}$$

Both arms see the same number of training trajectories and are scored on identical test data, so trajectory difficulty and set size difference out and the arms’ evaluation noise is correlated. The unpaired alternative did not survive contact with the data as it then stood: the unseen set’s variance differed from the seen set’s by -5.9% to +8.7% by split seed, a pre-fix figure the release does not reproduce. On the corrected bundles the imbalance is -6.1% to +10.7%, mean +1.4%, and the gate passes on that mean. Note that it passes on a statistic that is not the concern: the corrected per-seed spread is wider than the pre-fix one, so the risk the unpaired design runs is present in the release and pairing removes it by construction.

Gate	Caught
primitive collapse	<code>primitives_used = 1.0</code> , a “modular” model that is monolithic in disguise. No standalone artifact is released. Over every $K = 12$ modular model in the sweeps Table II reports, the minimum final-epoch <code>val_primitives_used</code> is 9.732 of 12 and exactly 16 sit below 11.0, all of them OakInk2’s; the withheld <code>pc_oakink</code> adds a seventeenth at 10.0. The $K = 16$ banks of §V-D and of §V-C’s pooled arm are outside that scope
informed coverage	held compositions absent from the informed set, which pins the penalty at zero by construction. The round-robin swap covers 5 of 5 on 18 of 20 confirmatory seeds and 16 of 20 affordance seeds, so the gate exits 1 on both; §V-B gives the failing rows
composition leak	48–69% of targets had their exact fine composition in the informed training set against 0% for naive, pre-fix; the released gate now reports 0.0% on both arms
DOF health	<code>thumb_mcp_abd</code> pinned at its limit in 100% of GRAB and DexYCB frames under an assumed abduction axis (pre-fix; the released gate now reports 40% and 33%, still above its own 25% threshold, so it exits 1; see §VI)
split balance	the risk that an unpaired design measures set difficulty. The gate tests the mean imbalance and passes at +1.4%; the per-seed spread it does not test is -6.1% to +10.7% (§III-C)

TABLE I
GATES, AND WHAT EACH CAUGHT

The fifth guards a risk the corrected data does not show; the other four each caught a defect in an earlier design.

Not every sweep is gated per seed. `oakink_class_v1`, `pc_easy_ratematched` and the 17 completing seeds of the confirmatory axis record a soundness block for each seed. The confirmatory axis’s first 52 predate that record, and standalone runs of the coverage and leak gates over a seed subset cover them instead. The split is deterministic given the seed, so those runs bound the same quantity. The coverage gate covers 37 of the 69, 20 as a standalone run and 17 through per-seed soundness blocks, leaving seeds 20 to 51 unrecorded. The leak gate covers all 70, both as a standalone artifact and in the per-seed soundness blocks of the rate-control sweep, which report 0.0% on both arms throughout.

The DOF gate tests per-DOF pinned fraction and variance directly. It replaces a clip-fraction statistic, and the reason matters: that statistic is defined on values *before* clamping, and an adapter that passes it post-clamp values returns ≈ 0 whatever the data does.

D. Predicting which axes are worth measuring

A full sweep costs hours of GPU per axis. A cheap statistic decides beforehand which axes repay that cost, and most axes do not.

The compositional penalty asks whether knowing the *pairing* helps beyond knowing each factor separately, which is an *interaction*. Two nested least-squares models per axis, on PCA-reduced windowed poses:

$$\text{additive pose} \sim \text{left} + \text{right} \quad \text{full pose} \sim \text{left} + \text{right} + \text{cell interaction} \quad R^2(\text{full}) - R^2(\text{additive})$$

with a permutation null that shuffles labels over whole *trajectories*, since 30 fps windows within one trajectory are near-duplicates.

A variance-explained statistic on the composition label alone does *not* work. Of the three real datasets scored on

η^2 , GRAB’s excess (+0.0538) and OakInk-Image’s (+0.0547) are indistinguishable, yet GRAB’s swept axis returns no compositional difficulty ($p = 0.494$) where OakInk-Image’s returns +0.00982. The OakInk-Image row is scored on the id-prefix axis §IV admits only for comparison, which is the axis whose η^2 was computed; it carries difficulty at $p = 1.1\text{e-}20$, so it serves the argument. The interaction term separates them, at $z = 8.3$ on the confirmatory axis against 3.9 on GRAB’s.

E. Retargeting to the Shadow Hand

Robot model. The robot is the Shadow Hand E3M5 model from MuJoCo Menagerie [18], in MuJoCo [17]. Anatomical DOF map to its joints by role, corroborated by an independent check, the two models’ ranges having been designed separately, so agreement is evidence the roles line up.² We transfer angles directly and clip them. Rescaling between the limit boxes would hide the saturation this is meant to measure.³

Contact and penetration. Fingers resting together in a closed grasp register contacts and are valid poses, so the report separates *touching* from *penetrating beyond a tolerance* and counts only the second. At zero configuration the model has zero contacts and every geometry margin is 0, so measured penetration reflects real geometry; that check is not in the release.

Abduction correction. Coordinate descent on the abduction joints only. Flexion carries what the motion *is*; abduction carries how far apart the fingers are held, which is what the index-against-middle failure needs more of. Each step tries every abduction joint at $\pm 1.5^\circ$ and keeps the move that most reduces the deepest penetration. Which sign spreads which finger is read from the model.⁴

IV. EXPERIMENTAL SETUP

Four public datasets (OakInk-Image [14], OakInk2 [15], DexYCB [16], GRAB [6]) plus two synthetic controls, one with planted via-point compositional difficulty and one purely interpolable. GRAB uses 8 of its 10 subject archives, which is how many downloaded. The other two are absent for no reason to do with the data. Budget 256 training trajectories per arm, 120 epochs, paired per-seed t -test with a sign test alongside, because a t -test alone cannot rule out a few extreme seeds carrying an effect. A result is declared only when both clear $\alpha = 0.05$. “Clears both declared tests” and “reaches neither” below refer to that pair. The banks of §V-C and §V-D use a different recipe: 200 epochs at stride 8 on full data.

²index_mcp_abd [-20, 20] against FFJ4 [-20, 20], thumb_mcp_abd [-10, 10] against THJ3 [-12, 12], thumb_ip_flex [-15, 80] against THJ1 [-15, 90]. Where they agree less well (thumb_mcp_flex [-10, 55] against THJ2 [-40, 40]) the per-joint clip rate reports it.

³The Menagerie model has 24 joints but 20 actuators, each finger’s two distal joints sharing a tendon, so writing qpos commands PIP and DIP independently where the physical hand cannot.

⁴Shadow’s abduction axes are [0,-1,0] for index and middle and [0,1,0] for ring and little, so a hand-written rule would be right for half the hand.

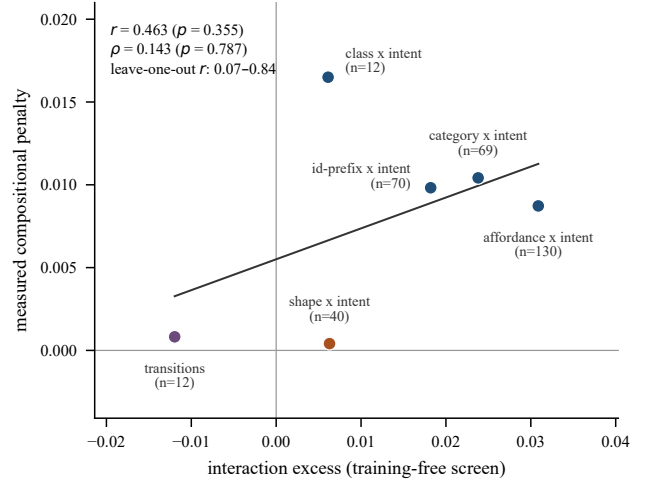


Fig. 2. A statistic computed on raw data factors, without training anything, against the compositional penalty each axis was measured to carry by a full paired sweep. Every axis with a measurable penalty lies right of zero and none lies left of it, so the statistic separates presence from absence; the scatter within the right-hand group is what it does not predict. DexYCB is absent because its sweep could not have measured what it claimed (§V-B).

The interpolable control returns no compositional advantage. At twenty seeds it gives +0.00154 with 12 of 20 ($p = 0.154$) at budget 256 and +0.00383 with 10 of 20 ($p = 0.438$) at budget 64 against the unmatched baseline. Against the rate-matched baseline of §III-B the figures are +0.00155 with 13 of 20 ($p = 0.090$) and +0.00324 with 11 of 20 ($p = 0.49$). Both arms land within noise of each other at both budgets.

We take composition axes from each dataset’s *own* taxonomy where one exists. OakInk ships object categories, functional classes and affordance attributes in its metadata, and all 100 object ids resolve. An earlier axis used the first character of the object id. That character records which sub-collection an object came from. Results on it are carried through the paper for comparison alone.

V. RESULTS

A. The modular arm’s advantage, and what rate matching does to it

What is measured. The advantage is how much smaller the modular arm’s compositional penalty is than the baseline’s, and how often. The difficulty that penalty acts on is a separate quantity, and we report the advantage first because it is the smaller and less certain of the two.

As shown in Table II, the modular arm’s advantage is positive on six of the seven interpretable axes, the six real ones and the synthetic control. It clears §IV’s declared pair of tests on four; a fifth, the class axis, clears the t -test alone. The two exceptions are the axes whose difficulty is itself indistinguishable from zero: GRAB’s shape axis, positive but at $p = 0.269$, and OakInk2’s, the one negative value in the table. The last column reads as the size of the problem and the second as how much of it the modular arm removes:

data / axis	advantage (wins)	d	p	difficulty
synthetic, via-point	+0.00491 (17/18)	1.13	0.0002	+0.00904
OakInk-Image, category $\times$ intent	+0.00397 (48/69)	0.41	0.0012	+0.01042
OakInk-Image, class $\times$ intent	+0.00873 (9/12)	0.65	0.046	+0.01650
OakInk-Image, affordance $\times$ intent (exploratory)	+0.00365 (86/130)	0.33	0.0002	+0.00873
OakInk-Image, id-prefix $\times$ intent (deprecated)	+0.01022 (60/70)	0.91	<0.0001	+0.00982
GRAB, shape $\times$ fine intent	+0.00305 (25/40)	0.18	0.269	+0.00041
OakInk2, annotated transitions	-0.00010 (4/12)	-0.05	0.860	+0.00082

TABLE II
THE MODULAR ADVANTAGE, AND THE DIFFICULTY IT ACTS ON

on the confirmatory axis, +0.00397 of +0.01042, or 38%. DexYCB’s sweep is omitted, for the reason §V-B gives.

On the confirmatory axis the modular arm wins 48 splits in 69, seven in ten, which is the number to weigh before the effect size. It recurs on the class axis at $d = 0.65$, swept after the screen mis-ranked it (§V-B), where three of twelve seeds cover 3 of 4 held compositions and the sign test is 9 of 12 at $p = 0.146$, short of §IV’s bar. We report that sign test because §VI treats the same statistic as disqualifying on the affordance axis.

Scale is the first thing to rule out. The baseline’s absolute error is about a third higher than the modular arm’s, so a larger absolute penalty could be an artifact of error scale. We normalize each seed’s penalty by its own naive error, which removes that confound. The advantage then survives on two of the three official axes. On the class axis it is +0.076 at 8 of 12 and clears neither test ($t p = 0.118$). On the affordance axis it is +0.045 at 77 of 130, clearing both of §IV’s tests ($t p = 0.004$, sign $p = 0.043$). On the confirmatory category axis it is +0.046 at 39 of 69, which clears the t -test ($p = 0.021$) and not the sign test ($p = 0.336$). Normalizing therefore costs the confirmatory axis its win rate, from 48 of 69 raw to 39 of 69, and by the rule §IV declares the normalized result stands on the t -test alone. Read all of these against §VI’s floor. On the raw penalty the unmatched arms differ by +0.00154 on a bundle with no compositional structure. On the normalized penalty they differ by +0.018, 39% of the +0.046 above. Real data bounds the metric harder than the synthetic control does. GRAB’s shape axis carries no measurable difficulty, +0.00041 at $p = 0.494$, and returns +0.116 normalized at 25 of 40 ($t p = 0.035$) against its unmatched baseline, above the +0.104 the confirmatory axis returns against the same arm; the withdrawn DexYCB axis returns +0.043, the confirmatory axis’s +0.046 to within noise. The metric’s maximum sits elsewhere and is real: the deprecated id-prefix axis gives +0.155 at 53 of 70, and that axis has both difficulty and the largest raw advantage. Wherever the modular arm reconstructs worse, dividing by naive error manufactures an advantage, so we treat the normalized metric as a robustness check that failed. The category axis carries the confirmatory claim and the affordance axis is exploratory, for reasons §VI

gives.

Rate matching. That normalization also decides how §III-B’s rate-matched comparison reads, and on the two metrics it reads differently. Take the scale-normalized penalty first, since its premise is most clearly met here: the rate-matched baseline reconstructs better than either other arm. On that metric, matching the arms on realized information rate appears to remove nothing: the advantage is +0.104 against the unmatched baseline and +0.111 against the matched one, 52 of 70 at $d = 0.54$, clearing both declared tests. That reading does not survive the negative control. On `pc_easy`, whose paired advantage is zero by construction, the same normalized contrast returns +0.099 at 18 of 20 and clears both tests, 90% of the figure just quoted. Normalization divides by each arm’s naive error, and on the control the matched baseline reconstructs 4.04 times better than the modular arm against 1.41 on real data, so dividing inverts the error-scale confound instead of removing it. The two floors are not subtractable across bundles with different ratios, which is the difficulty: this design cannot separate the advantage from the artifact.

On the raw penalty the same matching removes most of the effect, from +0.00693 at 50 of 70 wins and $d = 0.61$ to +0.00254 at 41 of 70 and $d = 0.23$, and the residual reaches neither declared test. The raw floor on the same control and arm is +0.00155, which is 61% of it. Both metrics therefore leave the rate-matched comparison inconclusive, one because the effect is gone and the other because the instrument produces almost the same effect on data that has none. We claim no modular advantage at matched rate.

Other decompositions. We ran two further decompositions against the same baseline on synthetic data. Both are taken off the shelf. `field` replaces the bank’s decoder with learned velocity fields under the same relaxed per-frame assignment, which is the switching-dynamical-system formulation of [3] and [4]; MANN selects and blends expert weights per frame instead of committing to one [5]. Each beats the parameter-matched baseline: +0.00491 for the bank over 18 seeds ($t p = 0.0002$), +0.00567 for `field` and +0.00968 for MANN over 8 ($p = 0.001$ and 0.010), against baseline penalties of +0.00904, +0.01018 and +0.00871 in the three sweeps that produced them.

Size does not explain those margins: `field` and MANN carry 1,919,832 parameters against the bank’s 2,295,727, and only the bank is parameter-matched to the baseline. The three pairwise contrasts between them are released and not argued from, since none separates at $n = 8$ and we ran no equivalence test. All three arms emit the same unpenalized per-frame assignment and carry §III-B’s rate confound identically. Our claim is about the measurement, so the architectures being prior art leaves it standing.

B. Which axes carry compositional difficulty

The screened axes. We screened eleven axes. The six in Table III are those whose sweeps are interpretable. The other five all score negative and are listed in `runs/axis_screen.json`. One of them, DexYCB’s

dataset, axis	left	right	excess (z)	difficulty
OakInk-Image, affordance $\times$ intent	0.054	0.045	+0.0309 (14.9)	+0.00873
OakInk-Image, category $\times$ intent	0.097	0.045	+0.0238 (8.3)	+0.01042
OakInk-Image, id-prefix (deprecated) $\times$ intent	0.028	0.045	+0.0182 (12.9)	+0.00982
GRAB, shape $\times$ fine intent	0.028	0.044	+0.0063 (3.9)	+0.00041
OakInk-Image, class $\times$ intent	0.022	0.045	+0.0061 (4.1)	+0.01650
OakInk2, annotated transitions	0.188	0.243	-0.0120 (-2.6)	+0.00082

TABLE III

THE INTERACTION SCREEN AGAINST THE MEASURED PENALTY

subject $\times$ shape, was also swept and is withdrawn below. Left and right give the two factors' separate η^2 , excess is §III-D's interaction term minus its permutation-null mean, with the z beside it, and the last column repeats Table II's difficulty so the screen and the sweep can be read against each other.

Informative factors are not enough on their own. OakInk2's two factors are the most informative in Table III, at 0.188 and 0.243 against OakInk-Image's 0.097 and 0.045, and pairing them adds nothing measurable; GRAB's swept axis is less informative on both and also null. The axes carrying difficulty share the interaction term and the informative ones lack it, so a practitioner choosing what to record should measure the interaction, which is the harder property to arrange and the one a factor-by-factor look at a dataset will miss.

Every axis carrying difficulty has positive interaction and none carries difficulty without it, but only one of the six could have shown otherwise: falsifying it needs negative interaction and measurable difficulty, and only OakInk2's is negative, at -0.0120. The withdrawn DexYCB axis is negative too and would have been a second test had its sweep been interpretable, so the relation rests on one informative observation and we state it that way throughout. Sufficiency is demonstrably false: GRAB's axis has interaction at $z = 3.9$ and returns +0.00041 at $p = 0.494$.

Four of the six return a penalty distinguishable from zero, all with positive excess: three at $p < 1e-19$ and the fourth, at twelve seeds, at $p = 0.0018$. GRAB's shape axis ($p = 0.494$) and OakInk2's ($p = 0.115$) are the two that do not. One of the four is the deprecated id-prefix axis, and setting it aside leaves five axes with three carrying difficulty, all from OakInk-Image; counts are given on the six so a reader can remove it, and the relation holds either way.

The screen does not predict magnitude. We swept six real axes before using it to choose one, and over the five that survive withdrawal the interaction excess tracks the baseline arm's penalty at $r = 0.843$, already not significant at that sample size ($p = 0.073$). We then swept the class axis on the screen's own recommendation, at twelve seeds, with the reading declared in advance: ranked fifth of six, it predicts a small penalty. It returned +0.01650, the largest difficulty we

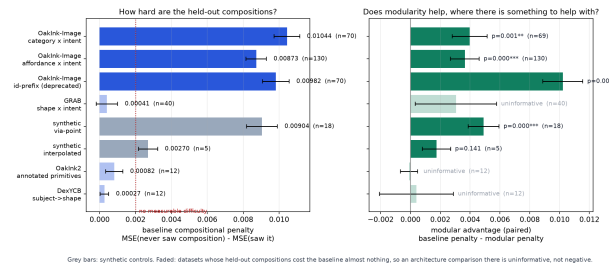


Fig. 3. Left, how much the held-out compositions cost a baseline that never saw them. Right, how much of that cost the modular arm removes, paired by seed. Faded rows are axes whose difficulty is itself indistinguishable from zero, where an architecture comparison is uninformative and not negative. Every value is read from the released run artifacts.

measured anywhere, and that one sweep took the association to $r = 0.463$ ($p = 0.355$). Removing the deprecated axis leaves $r = 0.868$ over four ($p = 0.132$) before that sweep and $r = 0.446$ over five ($p = 0.451$) after, so the collapse does not depend on it. The screen ranks axes badly, and we found that out by trusting it. Fig. 2 plots the two quantities: every axis with a measurable penalty lies right of zero and none lies left of it, and the vertical scatter inside that group is what the statistic does not predict. Fig. 3 gives the same six axes as measured quantities, difficulty beside advantage.

The one axis the statistic itself selected lies furthest from its own line, and the obvious escape does not hold. If the screen failed when the left factor is coarse, the id-prefix axis would fail too. It has the same five groups and is ranked correctly. We therefore use the screen to decide which axis to try first, and let the sweep say how hard it is.

A withdrawn axis. DexYCB's sweep is excluded because it could not have measured what it claimed. It ran at `min_chains = 1`, which disables the restriction, because all sixty transitions in its bundle fall in a single fine group. The coverage gate of §III-C then fails on all twelve seeds at both budgets, the informed arm reaching 1 or 2 of the 4 held compositions for a mean of 1.58.

The failure is one of breadth. Depth is fine: the compositions it reaches carry 5.0 to 16.0 examples, inside the 3.8 to 37.3 band the other axes sit in. However, a held composition the informed arm never sees is what Table I calls pinning the penalty at zero by construction. No configuration recovers it: every alternative raises the complaint that no transition appears in enough distinct groups, and the two single-factor coarsenings pass only by collapsing one factor to *any*. DexYCB's null belongs to the split, and the dataset is untested here.

We withhold `runs/pc_oakink` for the same reason: a five-seed sweep of OakInk-Image's fine object $\times$ intent axis, 345 occupied cells at 2.23 clips each, returning -0.00163 at budget 256 and -0.00343 at budget 64. Both are artifacts of a grid that thin.

Coverage. The remaining axes fall into three cases. We report them separately because our own gate transcripts for two of them covered ten seeds of sweeps that used forty and twelve, and the claim they supported stood for longer than

the evidence did. GRAB’s shape axis and OakInk2’s pass every row of their full sweeps, 80 rows and 24, which matters because they carry the nulls the necessity argument rests on. Four of GRAB’s 80 reach 5 of 6 and pass on a breadth rule of 0.8, where a row reaching 4 of 5 scores exactly 0.80 and fails; the same rule disqualifies rows below. Thinness costs a penalty its size, so the class axis is measured conservatively: it fails 6 of 24 rows on breadth, with seeds 1, 5 and 6 reaching 3 of 4 held compositions, and still returns +0.01650.

The other three fail a minority of rows on the same criterion: 4 of 40 confirmatory, 8 of 40 affordance, 4 of 40 id-prefix. Those rows still reach 4 of 5 held compositions at 8.0 to 28.0 examples each, against DexYCB’s 1 or 2 of 4. Excluding every seed flagged, by the standalone gate (8 and 13) or by a per-seed soundness block (59), moves the confirmatory advantage from +0.00397 over 69 to +0.00419 over 66 ($t p = 0.0010$), leaving the result where it was.

What produces it. We do not know which collection property produces the interaction, and two candidates fail. The first holds that OakInk-Image’s intents are more pose-relevant. Its own numbers refuse it: the intent main effect is 0.045 there and 0.044 on GRAB.

The second, that a crossed design produces it by recording one object under several named intents, fails harder. Over the eleven screened axes, grid occupancy and interaction excess correlate at $r = -0.095$ ($p = 0.78$). GRAB’s object $\times$ intent-class grid is 93% occupied with a significantly negative excess, both DexYCB axes are perfectly crossed and carry the two most negative scores, and OakInk-Image’s 86%-crossed fine axis is still unsplitable.

At $n = 11$ the interval on a correlation is wide, so neither candidate is refuted, but neither is what the data show. The axes returning a penalty all belong to one dataset whose collection procedure we cannot distinguish from the others on any property we measured.

C. Primitive identity

Reconstruction error says nothing about whether the bank’s units are units. The identity ratio is between-primitive over within-primitive variance in what a primitive produces, so above 1 means its output is determined more by which primitive it is than by what preceded it. Pooled over 13 runs carrying §III-B’s consistency term it is 1.841 ± 0.692 , with 11 of 13 above 1 ($t = 4.38$, $p = 0.0009$; sign test $p = 0.0225$) and a range of 0.87 to 2.84. It replicates on the dataset that carries the difficulty: those 13 runs are trained on OakInk2, which §V-A finds null, and at $n = 4$, where the sign test cannot clear 0.05 and is not quoted, four banks trained on OakInk-Image under an identical recipe give 1.848, all four above 1 ($t = 5.07$, $p = 0.015$).

Read that ratio as a target being met, since the consistency term optimizes exactly this quantity.

The ten banks trained without the term differ from the treatment arm in size or dataset, so we do not argue from them. Two smaller sweeps at $K = 12$ do hold size and dataset fixed and vary only the weight, which is what the ten do not; six of the ten are in fact those sweeps’ weight-zero arms, read

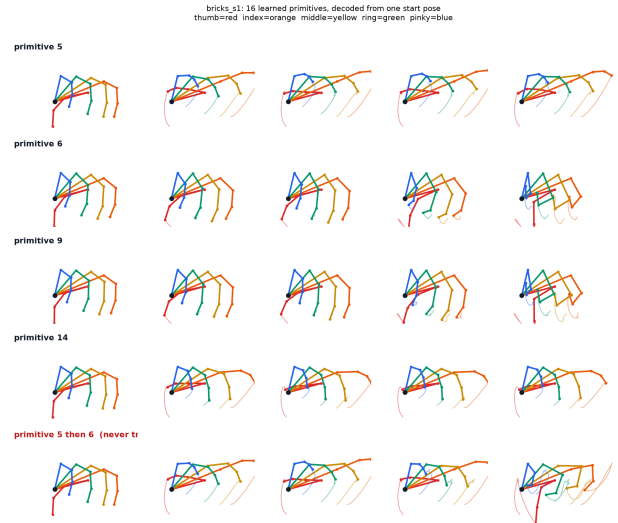


Fig. 4. Four of the sixteen primitives in one bank, each decoded from the same start pose and shown at five points through its window, with the trailing fingertip paths faded. The last row is one primitive followed by another, a pairing the bank never saw in training.

against the wrong comparison. Within their own sweeps both move the same way: mean identity 0.677 to 1.773 and 0.299 to 1.110 over three seeds each.

D. Transfer to the Shadow Hand

Setup. We attempted twenty-two comparisons between motion generated for unseen primitive pairs and for seen ones. A primitive pair is either seen or unseen and never both, so §III-C’s pairing is unavailable here. Grasp retention is scored by Fisher exact and the rest by Welch’s unpaired t -test. Five are undefined because both arms are identically zero: three where the abduction or penetration measure never varied, and two grasp-retention cells at hard where neither arm retained anything (0 of 23 and 0 of 40). Fisher’s exact test returns 1.0 there where Welch’s returns an undefined statistic, and we count them undefined because it is the data that leaves them so. Of the remaining seventeen, ten of them grasp-retention tests, four on abduction cost and three on penetration, sixteen fail to separate the arms and the seventeenth separates them in the opposite direction: abduction cost $p = 0.037$ with the unseen arm worse, 3.32° against 4.64° . Four tests at $\alpha = 0.05$ give at least one result about one time in five, which makes this consistent with no effect. It is still a contradiction, and we carry it forward as one.

Failure to reject is not evidence of similarity, so we test equivalence directly, and it does not survive its own sensitivity check. Pooled over five priors at 165 clips per arm, abduction cost is equivalent within 1° ($p = 0.008$) against a margin fixed in advance. But 60 of those clips per arm come from one prior whose correction never fired, all 120 of its values exactly 0.0. That block cannot fail a one-degree margin and it collapses the pooled variance. Dropping it leaves 105 clips per arm at $p = 0.124$, and dropping

bricks_s3, the prior carrying the reversal, from the full 165 moves the difference from -0.36° to -0.06° . Two blocks carry the test in opposite directions. The test needs more clips. At the pooled spread of 2.40° and a 0.36° difference over 165 clips, a one-degree margin needs 176 per arm for 80% power and 243 for 90% at $\alpha = 0.05$, by the normal-approximation TOST formula. On the 105 left without the degenerate block, spread 2.70° and difference 0.57° , it needs 488. Each bank’s seen arm is bounded by the primitive pairs its training data contains, at 19, 23, 23, 40 and 72. The last contributed 60 of its 72, so the 165 clips are $19 + 23 + 23 + 40 + 60$ and only four bounds bind. More clips means more banks. The other margins say little: across these bank-transfer runs penetration never exceeds 2.3% after correction, retention within ten points is not established at easy ($p = 0.069$) or medium ($p = 0.074$), and the pass at hard is vacuous at 3.3% against 0.0%.

The retargeting baseline. Direct retargeting of the first 40 trajectories per dataset clips 0.09%, 4.68% and 10.20% of DOF values on OakInk-Image, DexYCB and GRAB, and interpenetrates in 24.6%, 76.8% and 59.0% of frames. The abduction correction resolves 92%, 97% and 64% of that for 0.90° , 5.48° and 5.46° of deviation. GRAB is the residual case, and the 2.3% ceiling quoted above for the bank-transfer runs does not describe it: the three datasets are left with 2.1%, 2.1% and 21.2% of frames penetrating. The failure concentrates: 1,617 of 3,960 penetrating contacts on OakInk-Image are index-distal against middle-distal, and all six of its most frequent body pairs are index against middle, as are DexYCB’s. GRAB’s worst pair is little against ring, and GRAB’s thumb and little-finger channels arrive most clipped: 10.2% of its DOF values, THJ4 saturating in 87% of frames because the robot thumb cannot extend past zero where the human data goes negative. Joint travel the correction cannot reach, and GRAB is short of it; what the correction addresses is finger geometry.

At the task level, four modular banks against four monolithic priors, sampled unconditionally on a MuJoCo grasp-retention task, retain 28.1% and 27.7% at a pre-declared 480 clips per arm ($p = 0.94$), against 25.0% for uniform noise. Those monolithic priors carry 0.59 to 0.63 active units against the 0.996 to 1.000 of §III-B’s baselines, so the null is between a healthy bank and a partly collapsed latent. An exploratory run on the same eight models at 160 clips per arm gave 35.6% against 26.9% (Fisher $p = 0.117$); two further 160-clip files in the release are void, their monolithic arms having trained 10 to 13 epochs on a window-level latent, as `scripts/task_comparison.sh` records; it appears here because `scripts/task_confirm.sh` declared it would, and the confirmatory test exists precisely because that p value had been seen.

VI. LIMITATIONS

Section V-A’s two metrics disagree and the negative control disposes of both. Matching takes the raw effect from $+0.00693$ to $+0.00254$, past neither test and down to that arm’s floor. On the normalized penalty it appears to

remove nothing, but the same contrast on `pc_easy` returns $+0.099$. Normalizing was meant to correct for a baseline that reconstructs worse; the rate-matched baseline reconstructs better, so the correction runs the wrong way. No arm holds rate and reconstruction quality fixed at once, and no metric here separates the effect from the instrument. Intervening on the modular side is worse: charging the assignment its categorical KL zeroes that channel and raises reconstruction error by 57 to 72%, so its -0.00336 for `perframe_rate` against `modular_rate` (sign test $p = 0.041$, interpolable control) describes a model we broke.

Two sweeps of the same axis differ by more than either’s standard error, and this design separates neither cause: training three arms consumes the random stream differently from training two, and the sweeps use `min_per_composition` 4 and 5, which changes the informed set and leaves the naive set identical (§III-C). Read across sweeps, effect sizes carry that spread; §V-A’s rate-matched contrast sits within one sweep and is sound.

The paired penalty also returns a value where the truth is zero. `pc_easy` has no via-point offset, so its compositional penalty is zero by construction, yet in `runs/pc_easy_ratematched` the baseline arm reports $+0.00258$ ($p = 0.0002$) at budget 256 and $+0.00306$ ($p < 0.0001$) at budget 64 over twenty seeds. Taking the budget-256 floor throughout, that is 25% of the unmatched baseline’s difficulty in the re-sweep ($+0.01024$; Table II gives $+0.01042$ for the original) and 44% of the rate-matched baseline’s ($+0.00586$). Difficulty therefore reads partly off the measuring model, and on the arm this paper prefers the floor is nearly half of it. The same control bounds the advantage, and on one arm it bounds it away. On a bundle whose paired advantage is zero by construction, §IV’s unmatched arms differ by $+0.00154$ raw at budget 256, which is 39% of the confirmatory $+0.00397$, and by $+0.018$ normalized, 39% of the normalized $+0.046$. No headline should be read without these, and the rate-matched one does not survive them: against that baseline the raw floor is $+0.00155$, 61% of the whole effect, and the normalized floor is $+0.099$ at 18 of 20, which clears both declared tests.

We truncated one sweep and extended another after seeing its p value. The category sweep declared 70 seeds and stopped at 52. The missing 17 are null alone (-0.00046 , $t p = 0.86$, 11 of 17 still favoring modular), so the first 52 were the favorable half and pooling moves d from 0.60 to 0.41. The affordance axis has the opposite defect. Seventy seeds gave $+0.00270$ at sign $p = 0.120$, sixty more were run, and the pooled $+0.00365$ passes at sign $p = 0.0003$. Having looked survives completing the sweep, so that axis is exploratory and the confirmatory claim stays with the category axis.

The effect is small per split: the modular arm wins 48 of 69 seeds on the category axis and 86 of 130 on the affordance axis, helping on average and failing on roughly three splits in ten. The screen’s points are not independent either, since four of the six interpretable axes are OakInk-Image’s and share one intent factor on the same 770 trajectories, so six observations are closer to a one-dataset-against-two contrast

that two or three sweeps elsewhere would settle.

Transfer is kinematic throughout: contacts are read off after `mj_forward` places the model and nothing measures forces. Its baseline is weak in a specific way: since DexPilot [19] the retargeting work we are aware of operates in task space, and the widely used dex-retargeting implementation of that objective, released with AnyTeleop [21], carries no collision term. The 24–77% interpenetration figure therefore characterizes that baseline and not the field’s standard, so we make no claim about pipelines carrying a task-space objective. We ran one: in `runs/dexpilot_transfer.json` both arms penetrate in 100% of frames over four clips each, undefined for the reason §V-D gives for its other zero-versus-zero cells, and we report it because it cuts our way and cannot support us. Those banks were trained on OakInk2, which §V-A finds null.

Five further limitations are documented with the release, each with its artifact: the advantage does not grow as data shrinks, over four sweeps at budgets 64 and 256 with two reversing and none significant; contact does not explain the nulls, since GRAB’s 16 per-region channels leave the penalty at +0.00032 ($p = 0.813$); abduction is clipped everywhere, leaving 1 to 6 of 27 DOF pinned and wrist rotation unrepresented; both synthetic controls predate the split fixes, so their partitions are unreproducible; and `composition.json`’s interpenetration figure is a capsule proxy that does not track the mesh test even in rank.

The experiment that would settle this was not run. Sampling both priors unconditionally gives the architectures no channel through which to differ on grasp retention, which is why §V-D’s task-level comparison cannot see the effect. Seeing it needs each prior used as the action space for a policy that is trying to grasp, as LAMP [1] and DexSynRefine [7] both do for continuous latents. LAMP’s discrete VQ-VAE arm finds gesture jumps at codebook switches, but it quantizes the commanded pose where our bank keeps a continuous decoder and makes only the selection discrete, so that artifact is not entailed. A bank of this form against a parameter-matched continuous latent is what we would run next.

VII. CONCLUSION

Before spending GPU on a compositional axis, run the interaction statistic of §III-D over the raw pose arrays. It trains nothing and finishes in minutes, and a negative excess is a reason to look elsewhere: none of the axes we measured carried difficulty without positive interaction. A positive one licenses less than it appears to, because GRAB’s shape axis clears the same bar and returns +0.00041 at $p = 0.494$. Where difficulty is present the modular bank pays the smaller penalty. At matched rate we claim nothing for modular structure, because our own control manufactures most of the effect we would be claiming. A reader who wants that question settled should look to a bank used as a policy’s action space, which is where the architectures were proposed and where we did not test them. We stop short of a dataset

design rule. Every axis that returned a penalty belongs to one dataset, and across the eleven screened axes no collection property we measured predicts which.

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